Coastal Flooding on Transportation Infrastructures: A Systematic Review of Patterns, Impacts and Methodologies

Md Shah Mominul Islam Momin
Ingram School of Engineering
Texas State University
601 University Drive, San Marcos, Texas 78666
Email: shah_mominul_islam@txstate.edu
ORCID: 0009-0001-6872-4756

Eunsang Cho, Ph.D.*
Ingram School of Engineering
Texas State University
601 University Drive, San Marcos, Texas 78666
Email: eunsang.cho@txstate.edu
ORCID: 0000-0003-1841-6939

Eunsaem Cho, Ph.D.
Hydrological Sciences Laboratory,
NASA Goddard Space Flight Center
8800 Greenbelt Road, Greenbelt, MD 20771, USA
Earth System Science Interdisciplinary Center
University of Maryland
5825 University Research Court, College Park, MD 20740, USA
Email: eunsaem.cho@nasa.gov
ORCID: 0000-0002-3121-7494

Subasish Das, Ph.D.
Ingram School of Engineering
Texas State University
601 University Drive, San Marcos, Texas 78666
Email: subasish@txstate.edu
ORCID: 0000-0002-1671-2753

Submission Date: December 1st, 2025

*Corresponding author

ABSTRACT

Critical transportation infrastructure – central to societal well-being and economic activity - is facing increasingly severe risks from coastal flooding intensified by climate change and sea-level rise. Although the vulnerability of low-lying transportation networks is widely recognized, existing research on their complex and often systemic failure mechanisms remains fragmented across disparate disciplines, highlighting the need for a systematic synthesis to guide resilient decision-making. To address this gap, this study conducts a systematic literature review following Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, synthesizing findings from 83 peer-reviewed articles published between 2000 and 2025. Results reveal a clear evolution in the field, shifting from isolated single-hazard assessments toward more comprehensive evaluations of compound coastal events. However, three major deficiencies are identified. First, aligning with the IPCC AR6 hazard typology, the review identified a widespread omission of preconditioned events, specifically vertical land motion and subsidence. Second, a methodological disconnect exists between hydraulic modeling approaches and structural engineering analyses, limiting integrated risk characterization. Third, the literature exhibits strong model skew, with approximately 70% of studies exclusively targeting roadway systems, thereby overlooking cascading, system-level disruptions across critical intermodal nodes such as ports and railways. These findings underscore the need for next-generation modeling frameworks that integrate non-climatic drivers, capture multi-modal interdependencies, and support probabilistic, forward-looking, evidence-based adaptation strategies for coastal transportation resilience.

Keywords: Coastal Flooding, Transportation Infrastructure, Systematic Review, Compound Events, Vulnerability Assessment, Climate Change, Sea-Level Rise, Extreme Rainfall

INTRODUCTION

Coastal flooding, driven by the combined effects of sea-level rise, changing storm patterns, and intensifying coastal development, represents a growing and significant threat to urban environments worldwide (*1–7*). Among the various urban systems at risk, transportation infrastructure is uniquely vulnerable due to its extensive presence in low-lying coastal zones and its indispensable role in supporting economy, societal function, and emergency response (*2*, *8–11*). The failure of these critical networks under flood conditions can trigger cascading disruptions that extend far beyond the inundated areas, underscoring the urgent need for a comprehensive understanding of their vulnerabilities and the development of effective adaptation strategies (*8*, *9*, *12*). Past extreme events, such as Hurricane Katrina (2005), have repeatedly demonstrated the acute vulnerability and systemic interdependencies of coastal transportation networks, demanding a comprehensive and systematic analysis of the patterns, impacts, and methodologies used to study them.

While the body of research on coastal hazards is robust, existing literature reviews have tended to address the issue from broader or more segmented perspectives, creating a compelling need for a targeted synthesis that integrates these disparate themes. For instance, a significant portion of the literature is dedicated to understanding the physical mechanisms and modeling of compound flooding. These seminal works provide deep dives into the statistical methods for modeling driver dependence (e.g., copulas) (*13–16*), the application of various hydrodynamic models (e.g., Advanced Circulation Model: ADCIRC, Delft3D, LISFLOOD-FP) (*5*, *16–20*), and the complex interplay between drivers like storm surge, tides, and rainfall (*6*, *10*, *16*, *21*). Other reviews have concentrated on the evolution of large-scale flood hazard mapping or the understanding of extreme sea levels, providing critical context on hazard characterization but stopping short of a detailed analysis of specific infrastructure impacts (*22*, *23*). Simultaneously, another major branch of research reviews the vulnerability of critical infrastructure more generally, but often without a specific, in-depth focus on transportation. Some systematic overviews of coastal adaptation strategies, for instance, categorize transportation as one of several co-equal sectors alongside water, energy, and communications (*2*, *24*). While valuable for their broad scope, these works do not deeply investigate the unique functional, topological, and network-based vulnerabilities that define transportation systems (*8*). Other studies provide detailed frameworks for assessing interdependent infrastructure, offering crucial insights into how failures cascade between sectors—such as a power outage disabling an airport's control tower (*9*) or a water main break disrupting road access (*25*)—but their primary focus remains on the interdependencies themselves rather than the specific physical failure mechanisms of transportation assets. Similarly, reviews centered on specific sectors like coastal energy (*26*) or water infrastructure (*25*) logically concentrate on the risks to power plants or pipelines, leaving the direct and indirect consequences for transportation as a secondary consideration.

Furthermore, a distinct and substantial set of literature focuses specifically on road network vulnerability. These studies provide important bibliometric analyses, develop network analysis techniques to identify critical links (*8*, *12*, *27–29*), and create functional assessments like depth-disruption curves for traffic flow (*11*, *20*, *30*). However, this road-centric focus often means they do not integrate the full spectrum of coastal flood drivers beyond inundation, such as wave action (*17*) , salinity intrusion (*2*), or erosion (*31*). More importantly, this narrow lens often neglects the cascading impacts on interconnected, multimodal systems. The unique vulnerabilities of coastal rail lines to ballast washout (*2*), or the critical exposure of low-lying ports and airports that serve as essential hubs for trade and travel, are frequently unaddressed in these road-focused reviews (*9*,

24, *32*). This segmentation highlights a critical gap: the need for a review that synthesizes these disparate areas, connecting the physics of compound coastal hazards directly to the systemic, multimodal vulnerabilities of transportation networks.

This analysis of the existing literature reveals a clear and critical research gap: a lack of a single, systematic review dedicated exclusively to the multifaceted impacts of coastal flooding on transportation infrastructure. While prior work has laid the groundwork by examining individual components of the problem—such as specific flood drivers, general infrastructure resilience, or road network analysis—no study has yet provided a holistic synthesis that bridges these topics (*16*). There is a pressing need to connect the growing understanding of compound flood hazards directly to the specific vulnerabilities, systemic impacts, and assessment methodologies relevant to transportation systems, including roads, rail, ports, and airports (*12*, *33*, *34*).

Therefore, the primary objective of this systematic review is to address this gap by synthesizing and structuring this rapidly expanding field of study. To achieve this, the paper systematically examines several key themes, beginning with the patterns of coastal flooding, before analyzing the specific vulnerability and exposure of critical transportation assets. Subsequently, it explores the multifaceted impacts of these hazards and provides a critical analysis of the methodologies used for assessment. Ultimately, this review seeks to provide a foundational resource that identifies critical research gaps and opportunities, aiming to inform more integrated and resilient transportation planning in the face of escalating coastal flood risk.

METHODOLOGY

This paper employs a systematic literature review methodology to comprehensively identify, evaluate, and synthesize the existing body of research on the impacts of coastal flooding, especially on transportation infrastructure. The review process was structured according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines to ensure a transparent, replicable, and unbiased approach (see Figure 1). This structured methodology allows for a thorough examination of the drivers of coastal flooding, the vulnerability of transportation assets, documented impacts, and the evolution of assessment techniques. The process is divided into three main stages aligned with the PRISMA framework: (1) Identification, (2) Screening and Eligibility, and (3) Data Extraction and Synthesis.

The identification stage involved a comprehensive search of multiple academic databases to gather a wide array of potentially relevant publications. The primary databases used were Google Scholar, Web of Science, and Scopus, chosen for their extensive coverage of civil engineering, environmental science, and disaster risk literature. The search was performed using a Boolean search string that combined keywords related to the core topics of the review (Table 1). The search terms included but were not limited to:

Table 1. Search Keywords and Search Strategy

Search Concept	Keywords / Search Terms Used
Hazard	“Coastal flood*”, “Urban flood*”, “Climate change”, “Sea level rise”, “Extreme Rainfall”
Infrastructure	“Transportation”, “Transport infrastructure”, “Road*”, “Rail*” “Port*”, “Airport*”,
Impact/Outcome	“Vulnerability”, “Impact”, “Risk”
Search Logic	(Concept 1) AND/OR (Concept 2) AND/OR (Concept 3)

Note: The asterisk (*) denotes a truncation symbol used to retrieve all suffix variations of a root word (e.g., “flood*” captures “flood”, “floods”, “flooded”, and “flooding”).

The final search across all databases was concluded on May 31, 2025, ensuring the inclusion of the most recent publications. This initial search yielded a substantial 3073 number of publications for the screening phase.

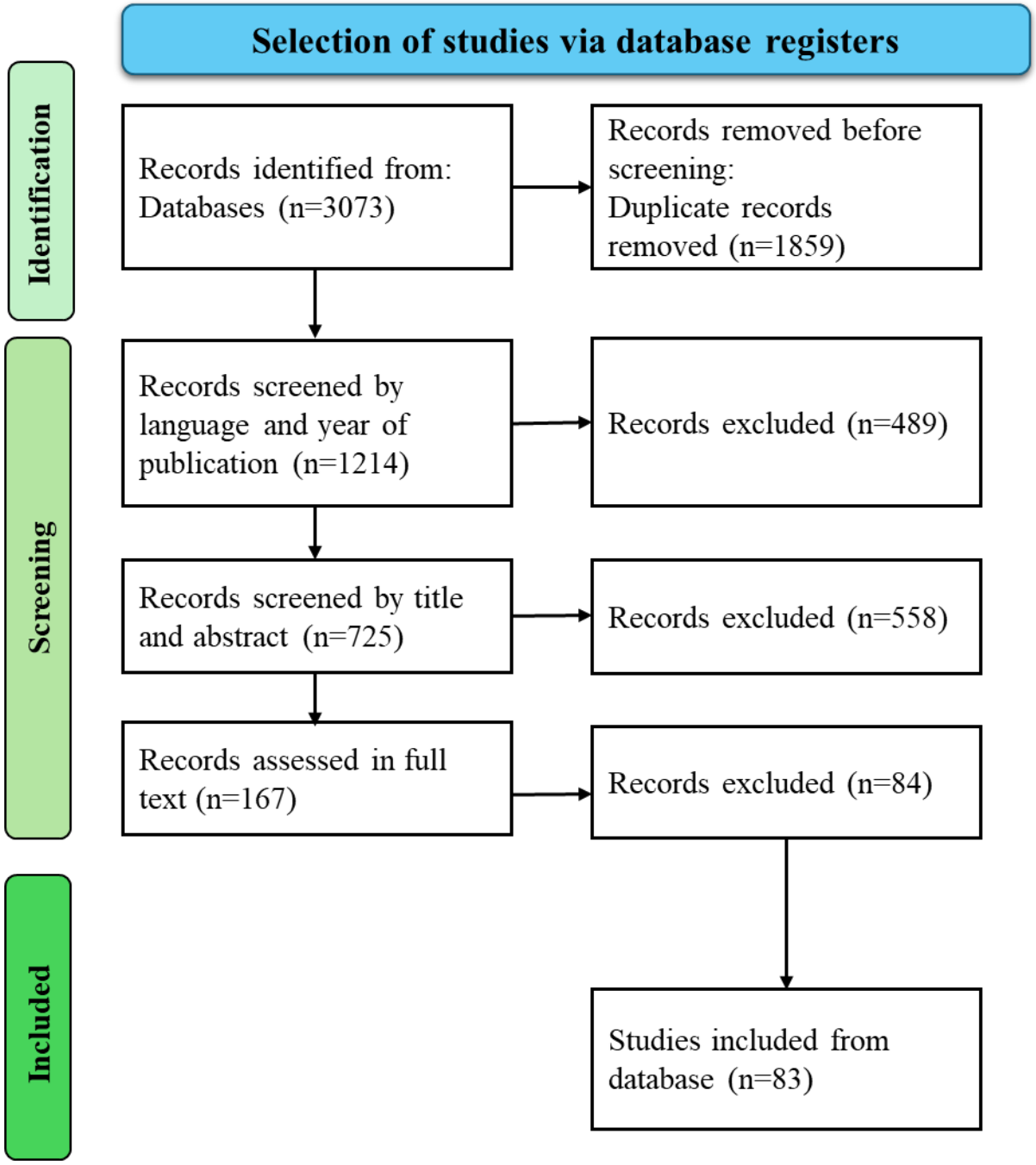

Figure 1. Flow Chart of the Process of Searching and Selecting Literature.

The screening and eligibility process was designed to systematically filter the initial pool of literature to only the most relevant studies through a multi-step approach. First, all identified records were compiled, and duplicate entries were removed using Zotero reference management

software to create a clean dataset for review. Following this initial step, the remaining unique records were filtered based on language and publication date; the scope was intentionally limited to English-language papers published between 2000 and 2025 to ensure a focus on contemporary research that reflects recent coastal flood trends and technological advancements. Subsequently, the titles and abstracts of the articles in this refined list were screened against predefined inclusion and exclusion criteria, a step that aimed to quickly remove publications clearly outside the scope of the review. The final stage of this process involved a thorough full-text assessment of all articles that passed the initial screening to determine their ultimate eligibility for inclusion in the qualitative synthesis. To ensure the reliability and accuracy of the selection process, a consensus-based screening methodology was adopted involving all four authors. The first author conducted the initial search and deduplication. Subsequently, the retrieved records were reviewed collectively by the research team. Rather than relying on statistical inter-rater reliability metrics (e.g., Cohen's *k*) derived from independent blind coding, the team prioritized a deliberative approach; uncertain or borderline cases were discussed in detail until a unanimous consensus was reached regarding their inclusion or exclusion based on the defined criteria. Inclusion criteria for the final selection required that a publication must:

- Focus primarily on the intersection of coastal flooding (including storm surge, SLR, and high tides) and transportation infrastructure (e.g., roads, rail, ports, airports).
- Provide a substantive discussion of flood drivers, infrastructure vulnerability, specific impacts, or assessment methodologies.
- Be an original research article, conference paper, or official technical report.

Exclusion criteria were applied to remove publications that:

- Only mentioned the keywords fleetingly without in-depth discussion.
- Focused exclusively on riverine (fluvial) or pluvial flooding without a direct coastal connection or compound context.

Definitions and Scope of Compound Events

To ensure theoretical consistency, this review adopts the compound event typology defined by the Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report (AR6). We specifically focused on events relevant to coastal transportation infrastructure, categorized as follows:

- **Multivariate Events:** The simultaneous occurrence of multiple climate drivers, most notably the coincidence of storm surge and fluvial discharge, or storm surge and heavy precipitation. This interaction is critical for coastal roadways where tailwater effects inhibit drainage (*35*).
- **Preconditioned Events:** Events where a climate driver (e.g., a storm tide) impacts an infrastructure system already primed by a background condition, such as long-term sea-level rise (SLR) or saturated soil conditions from antecedent rainfall (*35*).
- **Temporally Compounding Events:** Sequences of flood events occurring in close succession, preventing infrastructure recovery or repair between impacts (*35*).

A critical distinction was made regarding the physical boundaries of coastal flooding.

- **Inclusion:** We included studies analyzing estuarine and fluvial-coastal interactions. These regions represent the transition zone where riverine discharge interacts with tidal dynamics or storm surges, often resulting in compound water levels significantly higher than individual drivers would produce.

- **Exclusion:** We explicitly excluded inland pluvial flood studies where the flooding mechanism was solely precipitation-driven and hydrologically decoupled from oceanic influence. However, studies analyzing coastal pluvial flooding—where heavy precipitation occurs in coastal zones and drainage efficiency is modulated by tidal or surge levels (tailwater effects)—were retained, as these represent a critical sub-mechanism of compound coastal risk.

Descriptive Analysis of Selected Studies

To provide a quantitative overview of the field's evolution, we conducted a bibliometric analysis of the 83 selected articles. Figure 2 presents a circular network diagram mapping the relationships between publication year, flood drivers, transportation modes, and analysis methodologies. The visualization reveals distinct trends:

- **Temporal Trends:** There is a marked densification of connections post-2015, corresponding with the increased recognition of compound hazards.
- **Modal Distribution:** Road networks (orange nodes) dominate the literature, appearing in the majority of studies, whereas nodes for ports and airports show fewer connections, highlighting a gap in research of these sectors.
- **Drivers and Methods:** Storm Surge/Tidal (blue) and GIS/Spatial (red) remain the central hubs of research activity, though Compound drivers have shown stronger linkage strength in the last five years.

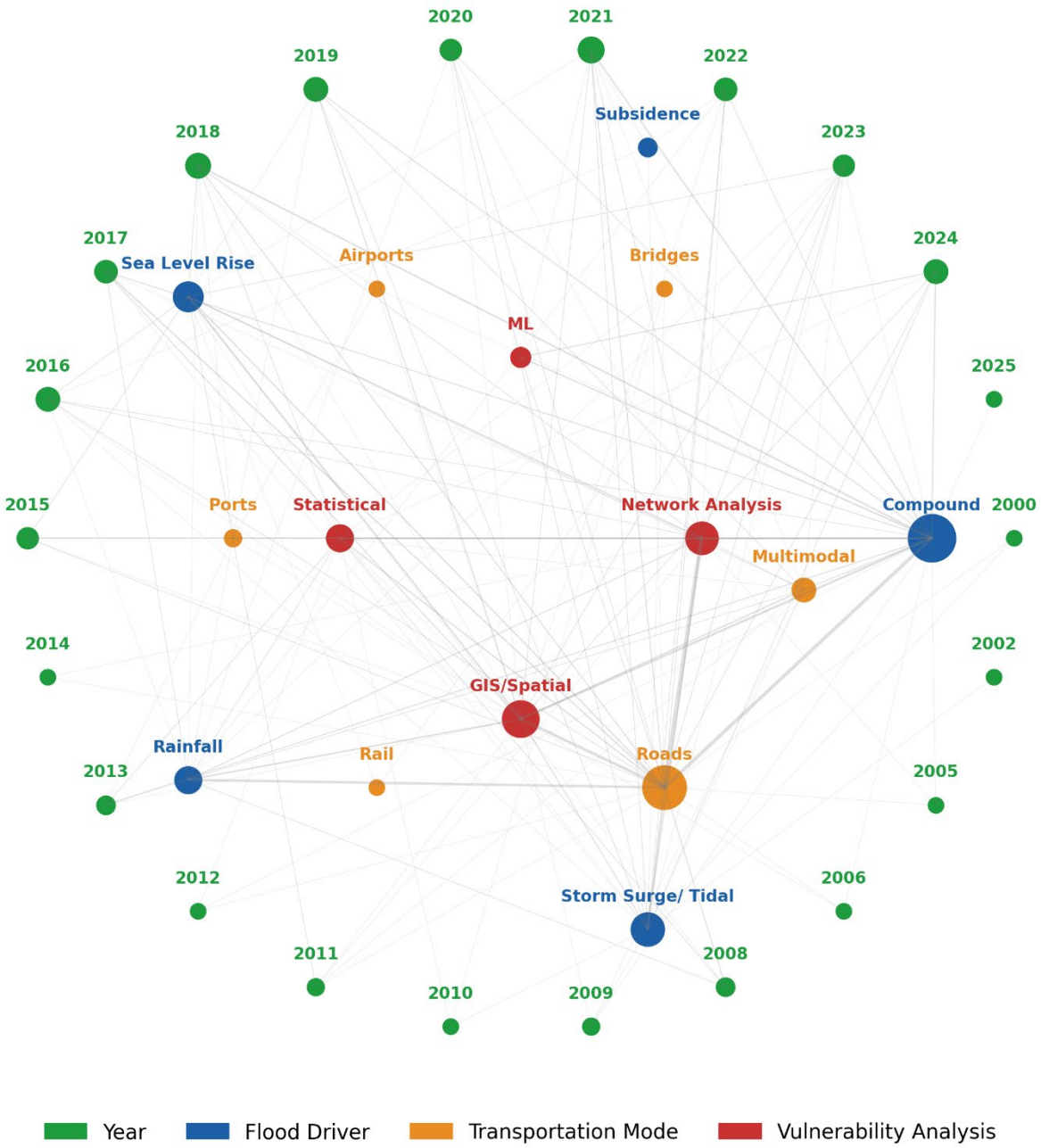

Figure 2. Circular network diagram illustrating the interconnectivity between publication years (green), flood drivers (blue), transportation modes (orange), and vulnerability analysis methodologies (red) within the reviewed literature. The radius of the circles represents the number of publications, comparatively.

For each article that met the eligibility criteria, key information was extracted and synthesized thematically. This qualitative analysis, guided by the conceptual framework illustrated in Figure 3, focused on categorizing findings according to five primary themes that form a logical and interconnected narrative. The schematic diagram depicts how climate-driven coastal flooding patterns influence vulnerability and exposure of transportation infrastructure, leading to direct and

indirect impacts, which are evaluated through various assessment methodologies. The integration of these themes highlights critical research gaps and future directions. The analysis begins with identifying the Patterns of coastal flooding, including the physical mechanisms and observed or projected trends under climate change. It then examines Vulnerability and Exposure, which details the methods and metrics used to assess the susceptibility of transportation assets to these hazards. Subsequently, the framework considers the documented Impacts of Coastal Flooding, ranging from direct physical damage to indirect service disruptions and economic losses. These impacts are understood through a critical analysis of the various assessment methodologies employed in literature, such as hydrodynamic modeling and GIS-based mapping. As depicted in the figure, the synthesis of these four interconnected areas allows for the identification of critical research gaps and opportunities, which represents the core contribution of this review. This thematic synthesis allowed for a structured narrative of the state-of-the-art, facilitating a clear discussion of key findings and future research directions.

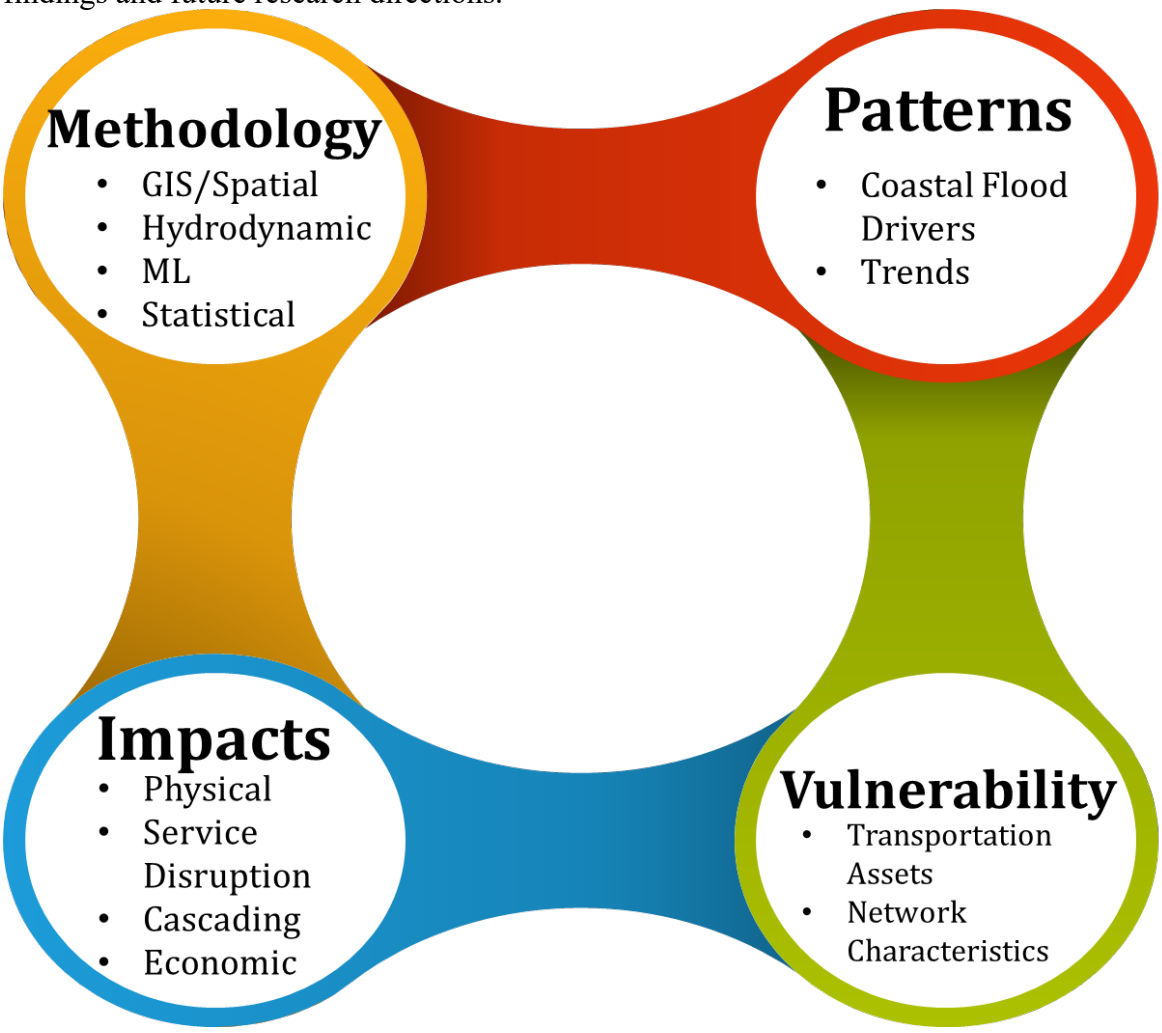

Figure 3. Conceptual framework demonstrating the four components of the study: patterns, vulnerability, methodology, and impacts.

PATTERNS OF COASTAL FLOODING HAZARDS

Coastal regions, globally critical for socio-economic activity and environmental services (*3*, *4*, *9*), face escalating flood vulnerability from both climatic shifts and increasing human development (*1*, *3*, *6*, *16*, *33*, *36–39*). Effectively assessing this threat requires a rigorous characterization of the hazard itself—a complex process driven by multiple interacting oceanic, atmospheric, and hydrological factors (*8*, *9*, *16*, *17*, *40*). While historical assessments often simplified this by analyzing these drivers in isolation (*16*, *41–44*), a growing body of research demonstrates that the most severe impacts arise from their synergistic combination, a phenomenon known as compound flooding (*16*, *45–47*). To fully grasp this evolving threat, the following analysis will unravel the primary drivers of coastal flooding, examine their powerful interactions, and synthesize the key trends that are fundamentally reshaping the hazard profile of our coastlines.

Drivers of Coastal Flooding

Coastal flooding is far from a singular, static phenomenon; instead, it is a complex and dynamic process driven by intricate interactions among multiple oceanic, atmospheric, and hydrological factors (*6*, *10*, *16*, *17*). Therefore, an effective assessment of the impact on critical infrastructure necessitates moving beyond single-driver analysis to a rigorous characterization of these complex interactions and the overarching trends that are reshaping the hazard landscape.

Sea Level Rise

The sustained, accelerating rise in global mean sea level, driven by thermal expansion of warming oceans and the melting of terrestrial ice, acts as a fundamental threat multiplier (*5*, *16*, *24*, *32*, *33*, *37*, *48*, *49*). This is a tangible, present-day outcome of climate change, directly increasing the frequency and severity of tidal and coastal flooding (*3*, *5*, *23*, *50–53*). SLR elevates the baseline upon which all other flood events occur, meaning that a given storm surge or high tide will reach higher and penetrate further inland than it would have in the past (*3*, *5*, *16*, *26*, *32*, *54*). Beyond simply raising the static level, SLR acts as a non-stationary driver that physically alters coastal hydrodynamics; specifically, it reduces the pressure gradients required to transport fluvial water to the ocean and facilitates greater upstream propagation of tides and waves (*37*). This dramatically increases the frequency of what were once considered rare, high-impact flood events (*3*, *5*, *37*, *55*). Indeed, the frequency of minor tidal flooding is accelerating nonlinearly because rising sea levels are evolving the nonlinear portion of a water level distribution against a fixed elevation threshold (*31*). Even linear SLR can result in exponential increases in flood frequency as water levels increasingly exceed local emergency thresholds (*31*). SLR contributes to increased total water levels when combined with other factors, and global mean sea level is projected to increase substantially by 2100 (*3*, *5*, *11*, *25*, *26*, *36*, *49*, *52*, *56*, *57*).

Storm Surge

An abnormal rise in sea level generated by a storm, over and above the predicted astronomical tide, is known as a storm surge (*13*, *16*). This critical driver, often caused by low atmospheric pressure and strong winds pushing water towards coastlines during storm events (*3*, *5*, *13*, *17*). The severity of the surge is often characterized by the "skew surge"—the difference between the observed peak water level and the predicted astronomical high tide (*58*). The coincidence of a major storm surge with a high astronomical tide is a classic formula for severe coastal flooding (*13*, *16*, *34*, *37*, *59*). Additionally, local bathymetry plays a significant role; large,

shallow continental shelves tend to produce higher surges than steep ones (*60*). Furthermore, the duration of the surge event is as critical as the peak elevation; longer-duration surges can generate greater volumes of backward flow into inland areas through drainage systems than shorter, higher-peak events (*18*). While SLR is the dominant long-term trend, the intensification of storm surges remains a potent threat, particularly as climate change is predicted to increase the intensity of hurricanes and cold-season storms (*32*, *61*).

Astronomical Tides and Wave Action

The predictable rise and fall of sea level from astronomical tides provides the foundational component of coastal water levels (*6*, *16*), creating an elevated baseline upon which storm-induced effects are superimposed, particularly during high-tide events (*37*). Adding a significant and highly dynamic component, wave action transfers wind energy to the sea surface, further elevating the total water level through run-up and overtopping (*5*, *17*). Notably, many large-scale flood studies have historically neglected this wave contribution, potentially underestimating total water levels and overall risk (*23*). Recent research indicates that in high-energy environments, wave runup is a dominant component, accounting for more than 50% of extreme total water levels (*62*). Future changes in ocean stratification and the loss of sea ice in polar regions may further exacerbate this by reducing frictional damping, potentially increasing tidal amplitudes and exposing coasts to increased wave energy (*40*, *59*).

Extreme Rainfall

Intense rainfall is a key hydro-meteorological driver causing both fluvial (river) and pluvial (rainfall-driven) flooding (*16*, *63*, *64*). This pluvial threat is particularly acute in urbanized areas, where impervious surfaces increase runoff that can overwhelm stormwater drainage networks (*9*, *30*, *41*, *64*). The hazard is significantly amplified when high coastal water levels from a storm surge or tide block drainage outfalls, creating a "backwater effect" that exacerbates inland flooding (*13*, *17*, *18*, *37*, *65*). This interaction is often non-linear; wind-driven surge can oppose gravity-driven runoff, altering hydrodynamics in "transition zones" where the combined flood level is often less than the linear summation of the two individual drivers (*60*).Furthermore, climate change is expected to increase the frequency and intensity of rainfall, heightening future flood risk (*30*, *37*). In dense coastal cities, this is further complicated by the "urban rain island effect," where waste heat accumulation can trigger increased local rainfall intensity (*7*).

Land Subsidence

Land subsidence, the downward vertical shift of the Earth's surface, is a critical factor that increases coastal flood vulnerability by lowering ground elevation, effectively magnifying the impact of SLR and making low-lying areas more susceptible to inundation (*5*, *9*, *66*, *67*). In many deltaic and coastal cities, subsidence is often the dominant contributor to relative sea-level change, impacting nearly one-fifth of the global coastal population and accelerating relative SLR by up to 10 times in some areas (*66*). This phenomenon is particularly extreme and well-documented along the U.S. Gulf Coast. Studies in this region identify land subsidence as a significant factor in accelerating Relative Sea Level Rise (RSLR) and elevating future storm surge risks (*68*). The underlying mechanisms vary by location. In Coastal Louisiana, subsidence is primarily linked to natural deltaic processes and hydrocarbon production, with geological modeling showing that subsurface fluid withdrawal can reactivate faults and cause significant land sinking (*69*). Along the Texas Gulf Coast, including the Houston metropolitan area, the dominant cause of subsidence

is rapid population growth and excessive groundwater withdrawal, though hydrocarbon extraction also contributes in some regions (*70*). The accelerated RSLR caused by these factors in major coastal transportation hubs significantly amplifies vulnerability, especially when combined with storm surge. With at least 15.6 million people in major coastal cities exposed to 0.5 meters of subsidence, omitting this factor leads to a significant underestimation of future risk, as projections of flood losses increase substantially when the combined effects of both SLR and land subsidence are considered (*1*).

Urbanization and Anthropogenic Activities

Rapid urban development and associated human activities significantly alter landscapes and exacerbate flood impacts (*7*, *41*, *63*). The proliferation of impervious surfaces, such as roads and buildings, reduces natural soil infiltration, leading to a sharp increase in the volume and velocity of surface runoff that can overwhelm urban drainage systems (*9*, *41*, *71*). In highly urbanized catchments, roofs alone can constitute 40-50% of impermeable surfaces, radically altering runoff coefficients (*8*). This problem is compounded by development patterns that concentrate populations and infrastructure in naturally flood-prone coastal plains (*10*, *63*). A specific anthropogenic driver is “storage encroachment,” where elevated house pads in new subdivisions displace floodplain storage, thereby increasing flood heights for surrounding areas (*10*). Furthermore, direct coastal modifications like land reclamation and dredging can degrade protective natural features, further increasing vulnerability (*2*). For example, channel deepening for navigation can facilitate the propagation of long waves inside estuaries, thereby increasing surge vulnerability (*72*).

While less common, other phenomena can cause catastrophic coastal flooding. Most notable are tsunamis, which are typically generated by seismic activity and can produce devastating inundation across entire ocean basins (*6*, *16*). However, a more insidious and often overlooked flood hazard in porous coastal regions is a rising groundwater table (*2*). As sea levels rise, the coastal water table also rises to maintain equilibrium, which can lead to "groundwater inundation"—flooding that emerges from the ground up, even far from the shoreline (*2*). This condition is exacerbated in low-lying cities where high marine water levels can reverse the hydraulic gradient, causing “backflow” through storm drains and “sunny day” flooding (*62*). A shallow water table also worsens pluvial flooding by inhibiting the ground's ability to absorb rainfall and impairing urban drainage systems (*10*, *41*); in some cases, saturation forces infiltration capacities to zero, converting all precipitation immediately into surface runoff (*18*). Neglecting these effects can lead to a severe underestimation of the total population and infrastructure at risk from coastal flooding.

Trends of Coastal Compound Flooding

The threat of coastal flooding is rapidly increasing, with a notable rise in academic publications on compound events since 2015 (Figure 2), underscoring the growing attention to this research area. The concept of "compound events," as recognized by the Intergovernmental Panel on Climate Change (IPCC), has spurred research into the combined impacts of multiple physical processes (*13*, *16*, *45*). Historically, coastal flooding was driven primarily by rare, extreme meteorological events; however, recent analyses reveal a fundamental shift toward a "cumulative hazard" regime characterized by frequent, lower-magnitude events driven by elevated sea-level baselines (*73*, *74*). Projections indicate that even without accounting for socio-economic

development, the frequency and physical extent of these hazards will escalate non-linearly due to the interplay of sea-level rise and changing storm climatology (*25*, *36*).

This escalation is driven by the increased frequency and severity of flooding events due to climate change and SLR (*3*, *5*, *6*, *24*, *31*, *37*, *48*, *51*, *75*). While the Global Mean Sea Level (GMSL) rose at a rate of 1.2–1.9 mm/year over the past century, satellite altimetry now indicates linear rates of approximately 3.3 mm/year, with future rates potentially reaching 16 mm/year by 2100 dependent on emissions pathways (*73*). This rising baseline has already fundamentally altered hydrodynamic forcing; for instance, changes in mean sea level have been shown to contribute nearly 90% to the difference in total water levels between historical storm events (*76*). Furthermore, thermodynamic effects described by the Clausius-Clapeyron relation are projected to increase rainfall intensity by 14–28% under warming scenarios of 2–4°C, significantly compounding the risk of pluvial flooding coinciding with storm tides (*9*). Neglecting these compound drivers—specifically the interaction between storm surge, heavy precipitation, and wave runup—can result in severe underestimations of flood hazard, with some studies showing a 60% underestimate of flooded areas when wave contributions are excluded (*22*, *54*).

Future projections suggest a drastic compression of return periods for extreme events. Events currently classified as 100-year floods are statistically projected to become annual or decadal occurrences by 2050 at the majority of tide gauge sites globally (*23*). Concurrently, "nuisance" or recurring tidal flooding is approaching saturation points along the U.S. East Coast, with projections indicating that high-tide flooding could occur nearly daily by the 2060s (*31*). This shift poses a heightened risk to coastal infrastructure, as joint hydrodynamic conditions that historically caused failure are projected to occur with increasing regularity, potentially rendering current defense standards obsolete (*73*, *76*).

Broader climate patterns, such as the El Niño-Southern Oscillation (ENSO), also modulate coastal flood risk on inter-annual to decadal timescales by altering sea levels, storm tracks, and the statistical dependence between drivers like rainfall and storm surge (*16*, *24*, *77*). Research indicates that specific ENSO phases, such as La Niña, drive a stronger dependence between storm surge and rainfall in regions like Australia, thereby exacerbating compound flood risk during these periods (*77*). Despite the growing understanding of these global drivers, local variability remains a critical factor. While less common in global assessments, vertical land motion (VLM) and land subsidence significantly magnify RSLR in many coastal-estuarine regions. In fact, peak subsidence velocities in major coastal cities often exceed the global mean sea-level rise rate, sometimes by two orders of magnitude in anthropogenic hotspots (*66*, *78*). This interaction creates a non-linear response in flood risk, where local micro-topography and subsidence rates can double the frequency of flooding with only minor increases in absolute water levels (*67*). Consequently, accurate future trending requires integrating these localized geological factors with broader climatic shifts to address the complex, cumulative impacts of compound flood-inducing factors.

VULNERABILITY AND EXPOSURE OF TRANSPORTATION INFRASTRUCTURE

Coastal transportation infrastructure is intrinsically vulnerable to flooding, which causes both direct physical damage and cascading disruptions to mobility, economic activity, and public safety (*2*, *9*, *16*, *26*, *33*). The escalating risks posed by climate change—including sea-level rise, increased storm intensity, and altered precipitation patterns—necessitate a comprehensive

understanding of these multifaceted vulnerabilities (*26*, *33*, *54*). The vulnerability of a transportation system is a multidimensional concept, determined not only by an asset's direct physical exposure but also by a confluence of geographic, infrastructural, and socioeconomic characteristics that dictate its susceptibility to damage and its capacity to recover (*2*, *9*, *29*, *33*). Therefore, achieving a holistic assessment requires analyzing the specific assets at risk, the topological structure of the network, and the drivers of compound flood events (*2*, *9*, *12*, *16*, *33*).

Critical Transportation Assets

The selection of critical transportation assets for flood vulnerability assessment is driven by their societal importance and inherent susceptibility to flood hazards (*9*). Assets such as roads, railways, ports, and drainage systems are prioritized due to their direct exposure, significant interconnectivity, and the potential for cascading failures (*2*, *8*, *9*, *79*). Since these infrastructures support essential services and trade, their disruption from flooding leads to widespread service interruptions and economic losses, justifying their prioritization in risk assessments (*8*, *9*, *30*).

Roads and Road Networks: Road and highway networks are exceptionally vulnerable to coastal flooding due to their low-lying location and designs based on outdated assumptions of a stable sea level, making them susceptible to sea-level rise and land subsidence (*2*, *6*, *9*, *10*, *17*, *24*, *27*, *29*, *31*, *31*, *33*, *37*, *51*, *53*, *66*, *73*, *75*, *80–83*). This vulnerability is intensified by recurrent nuisance flooding that causes regular disruptions and accelerates infrastructure decay (*2*, *16*, *31*, *51*, *84–87*). This nuisance flooding begins to cause disruptions at depths of just 10-13 cm, when water reaches the undercarriage of most passenger cars (*73*).The exposure is further exacerbated by the compounding nature of coastal flood hazards, where direct coastal inundation is often combined with pluvial flooding that overwhelms urban drainage systems (*10*, *13*, *15*, *16*, *18*, *21*, *30*, *34*, *37*, *41*, *64*, *71–73*, *83*, *88*). Specific operational thresholds define this vulnerability: road segments are often considered closed with as little as 7.6 cm (3 inches) of water (*89*), and many vehicles stall at 15 cm (*30*). A depth of 0.3 meters (30 cm) is widely considered the threshold for impassability, as it is sufficient to move an average car or cause vehicle instability through floating and sliding (*30*, *33*, *34*, *58*). The physical impacts of this inundation are severe, including pavement saturation, subgrade weakening, and erosion of the road base and bridge supports, with saltwater exposure adding a corrosive element that degrades materials (*2*, *20*, *90–92*). Critically, this vulnerability is systemic; the flooding of a few key links can trigger cascading failures that fragment the entire network, causing widespread disruption to commutes, supply chains, and emergency services (*8*, *9*, *12*, *27*, *29*, *33*, *65*, *80–82*).

Railways: Coastal rail lines, often critical for passenger transit and freight, are highly susceptible to flood-related damage due to their fixed location and sensitive components (*2*, *24*, *26*, *33*, *75*, *80*, *85*). Beyond structural threats, saltwater intrusion poses a critical electronic failure mechanism, causing technical faults and shutdowns in under-carriage electronics (*80*). Inundation and wave action can lead to ballast washout, which undermines the stability of the track bed and can cause track subsidence or depression (*2*, *33*, *85*). Such failures frequently lead to prolonged service disruptions and force difficult decisions regarding the high cost of either reinforcing coastal defenses or relocating infrastructure inland (*2*, *26*, *33*). Operational responses to these hazards can be graduated, as exemplified by the Adverse Weather Operations Protocol on the Dawlish rail line, which imposes a “Level 1” speed restriction, “Level 2” partial line closure, or “Level 3” full closure based on escalating hazard severity (*80*).

Bridges and Tunnels: As critical network chokepoints, bridges and tunnels face unique vulnerabilities (*2*, *29*, *33*). Bridges are threatened by pier scour from high-velocity flows and

reduced marine clearance from higher water levels (*2*, *9*, *20*, *33*, *85*). In addition, bridge superstructures are threatened by a combination of hydrodynamic forces, including buoyant force (lift), horizontal drag, and vertical impact (slamming) force from breaking waves, which can lead to deck unseating and catastrophic failure (*34*, *93*). Furthermore, the inundation of bridge approaches can sever network connections even if the main structure is undamaged, a common critical failure mode (*2*, *24*, *33*, *71*, *75*). Tunnels are highly susceptible to complete inundation via their low-lying entrances, which leads to the catastrophic failure of electrical, pumping, and ventilation systems and results in prolonged, costly disruptions (*2*, *33*, *85*). For underground rail, this ingress can lead to catastrophic costs; saltwater flood depths exceeding 5 meters in tunnels have been estimated to cause damage equivalent to 75% of the replacement cost for signals, power, and lighting systems (*94*).

Ports and Airports: As essential hubs for trade and travel, ports and airports are uniquely vulnerable due to their requisite location in low-lying coastal areas (*1*, *2*, *9*, *12*, *24*, *32*). For instance, a statewide assessment in California identified critical vulnerabilities in major airports and seaports under a compound scenario involving a 1.4-meter sea-level rise and a 100-year flood event (*24*). For airports, the failure criterion for runways is often the presence of standing water (*9*), though other climate mechanisms like runway (heat buckling) and reduced air density causing payload restrictions also disrupt operations (*95*). Floodwater can damage runways, terminal buildings, fuel depots, and critical supporting systems like backup electricity generators (*9*, *24*, *32*, *85*). The failure of these components can create a cascade of disruptions; for example, the loss of backup power at an airport can cripple air traffic control and terminal functions, demonstrating vulnerability through internal interdependencies (*9*, *32*). Similarly, the functionality of a seaport depends on its land-side connections; flooding of connecting road and rail lines can sever its logistical chain even if the port itself remains undamaged (*2*, *8*, *9*, *12*, *24*, *27*, *33*). Ports also face direct physical failure mechanisms, including the structural failure of quays, jetties, and breakwaters, port scouring and/or silting from altered coastal flows (*95*), and operational shutdowns from high winds (e.g., >18 m/s) or reduced bridge clearance for ships (*24*, *95*).

Network Characteristics and System Interdependencies

Transportation networks are inherently complex systems whose characteristics, including topology and interdependencies with other critical infrastructures, introduce systemic risks that can exacerbate flood impacts and trigger cascading failures (*8*, *9*, *11*, *12*, *16*, *29*, *30*, *65*, *80*, *82*). A network's physical layout significantly influences its vulnerability, as the failure of critical nodes like major intersections and bridges can cause severe accessibility reductions, especially when alternative routes are limited (*8*, *12*, *16*, *20*, *27*, *29*, *65*, *80–82*). Consequently, a network with low redundancy is highly susceptible to widespread failure from a localized disruption, a vulnerability that can be quantified using network analysis techniques to identify the most critical segments (*8*, *9*, *12*, *27*, *29*, *82*). Specifically, the techniques are used to measure route redundancy through metrics like the cyclomatic number and clustering coefficient (*96*), and to identify critical nodes and links by quantifying their betweenness centrality or closeness centrality (*8*, *96*).

This vulnerability is significantly amplified by the system's deep interdependence with other critical infrastructures like energy, water, and communications (*2*, *8*, *9*, *12*, *30*, *80*, *82*). A disruption in a supporting sector can propagate through physical, cyber, or logical linkages, triggering cascading failures that paralyze transportation even if the assets themselves are not directly inundated (*8*, *9*, *16*, *26*, *29*, *80*, *82*, *97*). These systemic impacts can be modeled as interregional ripple effects on the economy using Interregional Input-Output (IRIO) models that

are coupled with network disruption data (*98*). This web of dependencies becomes especially fragile during compound flood events, where multiple drivers like storm surge and heavy rainfall can simultaneously damage different types of infrastructure, increasing the probability of a system-wide collapse (*9*, *13*, *15*, *16*, *18*, *21*, *54*, *72*, *73*). Therefore, a comprehensive understanding of these interconnected topological and systemic vulnerabilities is crucial for developing effective resilience plans for emergency responders and planners (*8*, *9*, *29*, *51*, *90*).

IMPACTS OF COASTAL FLOODING ON TRANSPORTATION SYSTEMS

Coastal flooding imposes a cascade of impacts on transportation systems, from direct physical damage to systemic, network-wide disruptions (*33*, *51*). Initial structural damage from inundation precipitates immediate functional impairments like road closures and traffic congestion (*2*, *12*, *30*), which in turn generate significant economic losses from both direct repair costs and indirect business interruptions (*1*, *20*, *33*). As a foundational service, transportation failure can trigger cascading disruptions across other interdependent infrastructures, amplifying societal consequences and compromising public safety (*16*, *33*). A comprehensive understanding of this interconnected chain of impacts is therefore essential for developing effective adaptation strategies.

Physical Damage and Structural Degradation

The most direct consequence of coastal flooding is the physical damage to transportation assets, with roads, railways, and bridges being particularly vulnerable to inundation, erosion, and structural damage (*2*, *24*, *29*, *33*, *37*, *75*, *80*, *90*, *91*). Major roadways from Hawaii to Miami have demonstrated this vulnerability, and approximately 60,000 miles of U.S. coastal highways are currently exposed to periodic flooding (*31*, *33*, *50*, *85*). This damage is not limited to major storms; recurrent nuisance flooding also contributes significantly to the inundation, erosion, and structural degradation of low-lying road networks (*31*, *33*, *73*).

The physical impacts are severe: highway pavement integrity is compromised as runoff infiltrates road base layers (*30*, *33*, *88*, *92*), leading to saturation, subgrade weakening, and accelerated material degradation from saltwater exposure (*2*, *20*, *24*, *90*). High-velocity flows can cause severe erosion of the road base and bridge supports, while the weir-flow mechanism can undermine pavement from overtopping storm surge (*31*, *85*). However, damage magnitude is not solely a function of inundation depth; it is critically dependent on the velocity of the floodwaters (*24*). Recent studies emphasize that floodwater velocity often carries equal or greater significance than inundation depth alone, with empirical analyses showing that flow velocity is a statistically significant factor in road infrastructure damage (*20*, *99*). Not only do these high-velocity flows drive erosion, but the flow itself can trigger component- and system-level failures of transportation assets due to the high lateral load arising from dynamic water movement (*73*). Vulnerability assessments are, therefore, increasingly framed by the infrastructure's susceptibility to this hydrological load (*99*).

Key mechanistic pathways for this hydrodynamic failure include hydrodynamic drag, lift, and buoyancy on superstructures. Drag represents the intense horizontal force component in the direction of the flow, while lift and buoyancy represent upward vertical forces that can destabilize or float a superstructure off its supports (*93*). These forces are particularly dangerous for bridge decks; research indicates that when the peak vertical wave load (lift and buoyancy) exceeds the dead weight of the bridge deck, deck unseating occurs, disengaging the bearings and allowing horizontal drag forces to displace the deck transversely (*93*). Specific velocity thresholds have

been identified for these structural risks; flows exceeding 2.0 m/s are likely to cause severe structural damage to pavements and foundations (*20*), and transportation assets are destabilized when these hydrodynamic driving forces exceed the resisting forces of self-weight and friction (*34*).

Moreover, high-velocity surges introduce secondary hazards through debris impact and wave slam. Fast-moving flood water transports large objects—such as logs, broken structures, and even washed-away vehicles—which impart high-magnitude impulsive loading when striking vulnerable elements like bridge piers or abutments (*73*). These floating objects act as a battering ram against structures (*24*), and research has demonstrated how washed-away vehicles can become hazards themselves, causing blockage and additional structural damage at bridge sites (*30*). Additionally, breaking waves apply short-duration, extremely high-pressure forces—known as wave slam—to the underside of bridge decks or barriers. This phenomenon occurs when water traps pockets of air between the wave surface and the bottom of the bridge deck, causing a sudden, violent impact force (*93*).

The magnitude of hydrodynamic forces during extreme events offers clear case studies of system failure. Post-event analysis of Hurricane Katrina revealed that bridge damage across the Gulf Coast region was attributed not only to scour, but also to storm surge, wind, impact by debris and barges, and wave forces on bridge decks (*100*, *101*). Specifically, the overall repair or replacement cost for the damaged bridges was estimated at over $1 billion (*101*). The documented failures of bridges in Mississippi and Louisiana confirmed that the primary failure mechanisms involved lateral loading and wave slam on superstructures, underscoring the critical need to account for dynamic forces in risk assessment (*100*).

Coastal rail lines are similarly susceptible, with inundation leading to ballast washout that undermines track stability, often necessitating costly repairs or relocation, as exemplified by the 2014 storm in Dawlish, England (*2*, *24*, *75*). Projections show these risks are increasing significantly; a 1.4-meter sea-level rise is projected to double the length of at-risk highways in California, while nuisance flooding is expected to become a near-daily occurrence by 2065 at sites along the U.S. East Coast (*24*, *31*).

Service Disruptions and Network Functionality

Beyond physical damage, the most immediate impact of flooding is the widespread disruption of transportation services, manifesting as road closures, traffic delays, re-routing, and congestion that severely impede mobility (*2*, *8*, *12*, *30*, *31*, *33*, *51*, *65*, *84*, *88*). Road closures are a primary consequence of inundation, a disruption now commonly caused by nuisance high-tide flooding in addition to major storms (*16*, *30*, *31*, *33*, *51*, *73*, *84*). Quantifiable disruption thresholds indicate that vehicle speeds reduce non-linearly as water rises; a flood depth of just 10 cm can significantly reduce safe driving speeds, while 30 cm is identified as the critical threshold where roads become effectively impassable for passenger vehicles (*30*). Even in the absence of storm events, high marine water levels can hydraulically lock drainage systems, causing tailwater effects that result in street flooding and hazardous driving conditions (*18*, *62*).

These closures force traffic diversions, leading to severe congestion, increased travel times, and a measurable reduction in city-wide accessibility (*2*, *8*, *12*, *27*, *29*, *33*, *51*, *65*, *80*). Network centrality analysis reveals a disparity in this disruption; flood events cause a significantly higher functional drop for long-distance trips (>10 km) compared to local trips (*8*). Furthermore, while many zones may experience minimal effects, specific isolated communities can face up to a 100% reduction in accessibility, effectively severing them from the network (*50*). Assessments have

shown that flood events can nearly double traffic delays and lost trips (*12*, *33*, *50*, *53*, *102*). During these events, traffic volumes on access-controlled interstates often decrease while volumes on principal arterials increase, implying that travelers divert to lower-capacity roads to maintain routing flexibility, thereby exacerbating local congestion (*51*, *84*). The cumulative effect is often a fragmented transportation network where inundated segments alter travel patterns, compromise connectivity, and diminish the reliability of the entire system (*8*, *29*, *33*, *65*).

Cascading and System-Wide Impacts

The failure of transportation infrastructure during a flood rarely occurs in isolation; due to the high degree of interdependency among urban systems, a disruption in the transportation network can trigger cascading failures across other critical sectors (*9*, *12*, *16*, *26*, *33*, *80*, *82*). The inability to access facilities due to flooded roads can cripple the operations of power plants, water treatment facilities, and communication hubs, even if those facilities are not directly inundated (*8*, *9*, *24*, *26*, *33*, *71*, *82*). For example, a risk assessment in Saint Lucia showed how flooded airport access roads could lead to a failure of backup power systems, shutting down air traffic control and severely impacting the national economy (*9*). These interdependencies mean that failures in non-transport sectors can also propagate; for instance, failures in water infrastructure have been shown to trigger disruptions in electricity and transport, with over 80% of economic losses in one case study being attributed to indirect industry losses from interrupted water services (*9*, *25*, *26*). Additionally, logistical chains are uniquely vulnerable; even if a port facility is protected, the flooding of connecting road and rail lines can sever land-side connections, potentially halting national supply chains (*24*). The interruption of transportation networks also critically impedes emergency services, as was demonstrated during Hurricane Sandy when power failures necessitated the evacuation of a major hospital (*9*, *26*). Moreover, cascading impacts also extend to social infrastructure; the flooding of schools forces the displacement of students to surrounding institutions, introducing secondary social hazards to the community (*71*). Finally, a systemic feedback loop exists where subpar pre-disaster maintenance increases vulnerability to flooding, while the flooding itself accelerates the physical deterioration of road bases, further increasing future network instability (*92*).

Economic and Socioeconomic Consequences

The economic consequences of transportation disruptions from coastal flooding are substantial and multifaceted.

Direct Costs: Direct costs from coastal flooding involve the tangible, monetary losses from repairing and replacing physically damaged transportation assets (*19*, *20*, *24*, *30*, *90*, *92*). This damage includes the deterioration of pavement, erosion of the road base, and scour damage to roads and bridges (*2*, *31*, *80*, *90*, *92*). The financial impact is staggering; total global flood losses since 1980 have exceeded $1 trillion, with single events like Hurricane Harvey causing an estimated $152.5 billion in damages (*16*, *19*). For the Metropolitan East Coast of the United States specifically, a global sea-level rise of 1 meter is projected to triple the annualized costs of infrastructure damage due to flooding, rising to USD 1.5 billion per year by 2100 (*31*). Even at a local level, costs are substantial; California's Department of Transportation, for instance, spent $8 million on preventative measures for one section of highway after a lengthy closure (*2*). Similarly, Expected Annualized Losses (EAL) for the Boston rail system have already more than doubled since 2008 to $24.4 million per year due to minor sea-level rise (*94*). Projections indicate a dramatic rise in future losses, with the number of U.S. properties at substantial risk expected to

increase to 16.2 million by 2050, and average global flood losses in major coastal cities projected to exceed $50 billion annually by the same year (*1*, *34*, *64*, *73*). In California alone, a sea-level rise of 1.4 m is projected to expose a wide range of critical infrastructure to 100-year flood events, including 3,500 miles of roadways and 280 miles of railways (*24*).

Indirect Losses: Indirect losses from coastal flooding are often far greater than direct repair costs and include expenses from traffic delays, increased fuel consumption, negative impacts on tourism, and widespread business interruptions (*2*, *9*, *20*, *30*, *31*, *33*, *51*, *80*, *82*, *92*). A Study in Galicia, Spain indicate that indirect losses to industries caused by infrastructure service interruptions can account for more than 80% of total flood losses (*25*). Transportation disruptions are a major contributor, with passenger and cargo delay costs frequently surpassing direct damage costs (*20*, *33*, *99*). For example, nuisance flooding on the U.S. East Coast alone caused over 100 million vehicle-hours of delay annually, a figure projected to increase to 3.4 billion by 2100 under an intermediate sea-level rise scenario (*31*). Projections for coastal rail infrastructure in the UK suggest that line restrictions could increase from a historical average of 10 days to nearly 120 days per year by 2100, severely impacting labor mobility (*103*). The impact on tourism can be particularly severe when access to airports, ports, and recreational areas is compromised (*9*, *33*).

Socioeconomic and Environmental Losses: Coastal flooding inflicts significant socioeconomic and environmental losses, profoundly disrupting human communities and natural ecosystems. Socially, billions of people in coastal zones face heightened flood risk, with vulnerable populations, including low-income communities and communities of color, being disproportionately impacted due to fewer resources for recovery, which exacerbates existing inequalities (*2*, *16*, *21*, *24*, *29*, *33*, *55*, *63*, *104*, *105*). Hurricane Katrina studies illuminated systemic vulnerabilities within the transportation sector, particularly regarding evacuation planning and environmental justice. Failures in evacuation planning were noted for regions with low car ownership rates, highlighting a chronic neglect of foreseen risks and a critical lack of coordination in utilizing existing public transportation assets (*106*). This event demonstrated that effective disaster risk reduction requires establishing more resilient communities by addressing vulnerabilities influenced by the built environment (*107*). The jeopardy to critical infrastructure, including water, healthcare, and transportation systems, leads to interruptions in essential services, the isolation of communities, and impediments to disaster recovery (*2*, *24*, *25*, *29*, *92*). Environmentally, coastal flooding contributes to the degradation of natural systems through the submergence of low-lying lands, accelerated coastal erosion, and the loss of vital natural buffers like wetlands (*2*, *16*, *24*, *90*, *105*). Furthermore, saltwater intrusion into freshwater resources and an increased risk of toxic chemical exposure from hazardous materials sites pose severe consequences for ecosystems and public health (*2*, *24*).This highlights that the most significant risks often lie at the intersection of multiple systems, and a failure in transportation can be the initiating event in a much broader, system-wide crisis, severely hampering emergency response and recovery efforts (*5*, *9*, *16*, *26*, *29*, *33*, *34*, *80*, *82*).

METHODOLOGIES FOR ASSESSING COASTAL FLOOD RISK

The assessment of coastal flood risk to transportation systems is a complex, multi-disciplinary field that leverages a variety of methodologies to quantify hazards, vulnerabilities, and potential impacts (*2*). This literature demonstrates a clear evolution from single-hazard, deterministic assessments toward holistic frameworks that incorporate the amplified risk of compound events (*16*, *54*), cascading failures within interdependent infrastructure systems (*9*), and the deep probabilistic uncertainties inherent in future climate projections (*36*). These methods are

broadly classified into five key areas: (i) GIS-based spatial analysis, (ii) hydrologic and hydrodynamic modeling, (iii) network analysis and accessibility-based modeling, (iv) statistical and machine learning approaches, and (v) integrated frameworks that combine these techniques for comprehensive risk assessment.

Spatial Analysis and GIS-based Approaches

Geographic Information Systems (GIS) are an essential framework for coastal flood risk assessment, serving as the primary platform for integrating, analyzing, and visualizing diverse spatial datasets (*7*, *10*, *24*, *30*, *41*, *63*, *65*, *71*, *75*, *81*, *88*, *108*). The fundamental application of GIS is conducting exposure analyses by overlaying flood hazard layers with asset inventories—including infrastructure, buildings, and demographic data—to understand where hazards and vulnerable populations intersect (*10*, *12*, *24*, *29*, *57*, *80*, *81*).

A critical prerequisite for these assessments is a high-resolution Digital Elevation Model (DEM), often derived from LiDAR, which provides the detailed topographic data necessary for accurate inundation mapping (*5*, *12*, *18*, *19*, *27*, *34*, *36*, *37*, *58*, *59*, *72*, *75*, *88*, *90*, *108*). While these DEMs enable techniques like the static "bathtub model," this approach is widely criticized for its failure to account for hydraulic connectivity, often leading to inaccurate flood extent predictions (*3*, *5*, *22*, *37*, *59*, *108*).

Beyond simple inundation mapping, GIS serves as the analytical engine for more sophisticated vulnerability assessments using Multi-Criteria Decision Analysis (MCDA) techniques like the Analytic Hierarchy Process (AHP) (*7*, *29*, *63*, *71*, *82*, *109*). This framework allows for the systematic weighting of various vulnerability indicators—such as population density, road density, and land use—which are then aggregated to produce composite flood risk maps (*29*, *63*, *71*, *104*). The result is a powerful, spatially explicit tool that helps decision-makers identify high-risk areas and prioritize mitigation actions (*29*, *41*, *63*).

Hydrologic and Hydrodynamic Modeling

To simulate the complex physical processes of flooding, researchers utilize integrated numerical models, often in a framework where hydrological models (e.g., Hydrologic Engineering Center – Hydrologic Modeling System (HEC-HMS)) simulate rainfall-runoff to provide upstream boundary conditions for a “recipient” hydrodynamic model (*9*, *16*, *21*, *56*). These hydrodynamic models solve the governing shallow water equations to simulate flood depth, velocity, and inundation extent (*5*, *37*, *58*). The literature shows a clear progression from simple one-dimensional (1D) models toward more robust coupled 1D-2D models, which leverage the efficiency of 1D for in-channel flow while using a 2D model for detailed simulation of overland flow across complex urban floodplains (*21*, *34*, *37*, *56*). These dynamic models are a significant improvement over simplistic static “bathtub” methods, which neglect hydrodynamic processes and tend to overestimate inundation (*22*, *56*, *59*).

A wide variety of models (see Table 1) are cited for their application in coastal flood risk assessment for transportation:

1 **Table 2. Key Hydrodynamic Models Used in Coastal Flood Risk Assessment**

Model	Model Type	Brief Methodology	Source
HEC-RAS	Hydraulic Model (1D-2D Coupled)	Hydrologic Engineering Center's River Analysis System (HEC-RAS) combines one-dimensional (1D) modeling for efficient in-channel flow with two-dimensional (2D) modeling for detailed overland flow. This coupled approach provides a more detailed and accurate representation of compound flooding scenarios where multiple drivers, such as river discharge and coastal storm surge, interact in complex urban and coastal floodplains.	(*18*, *37*, *73*, *82*)
LISFLOOD-FP	2D Raster-Based Hydrodynamic Model	LISFLOOD-FP solves a simplified version of the shallow water equations and is computationally efficient, making it suitable for large-scale simulations over entire regions. It is widely used to simulate coastal flood events and assess inundation extent.	(*19*, *22*, *58*, *90*)
ADCIRC and SWAN	Coupled Hydrodynamic and Wave Models	The Advanced Circulation (ADCIRC) model is a hydrodynamic model for tides and storm surges. It is frequently coupled with the Simulating Waves Nearshore (SWAN) model to incorporate wave-driven processes like wave setup and runup. In this coupled system, SWAN provides wave data to ADCIRC, which in turn provides updated water levels and currents back to SWAN, creating a robust feedback loop for comprehensive hazard assessment.	(*16*, *17*, *37*, *83*, *90*)
MIKE Suite (DHI)	Integrated 1D-2D Hydrodynamic Modeling Suite	This professional software package includes models like MIKE 21 and MIKE FLOOD, which are used to combine 1D and 2D models for detailed and integrated flood simulations in coastal and urban engineering studies.	(*21*, *37*, *82*)
GCOM2D	2D Hydrodynamic Coastal Ocean Model	Generalized Coastal Ocean Model – 2D (GCOM2D) is designed to provide an integrated simulation of the various physical processes contributing to coastal flooding by incorporating the combined effects of tides, storm surge, freshwater runoff from tributaries, and forcing from wave breaking into a single simulation.	(*17*)
SELFE and SCHISM	Unstructured Grid Hydrodynamic Models	Semi-implicit Eulerian-Lagrangian Finite Element model (SELFE) and Semi-implicit Cross-scale Hydroscience Integrated System Model (SCHISM) open-source models use unstructured grids, making them effective for simulating inundation across complex coastal geometries, from the deep ocean to inland creeks. They are often used to assess compound flooding driven by the interaction of river flow and storm surge, with SCHISM being capable of handling complex, multi-driver events through 3D baroclinic modeling.	(*16*, *92*)
Delft3D Suite	Integrated Hydrodynamic Modeling Suite	A flexible modeling suite that includes components like Delft3D-FLOW and Delft3D-WAVE. It is used to simulate compound flooding by linking riverine, pluvial, and coastal processes. Its Flexible Mesh (FM) capabilities are particularly useful for creating high-resolution simulations of complex coastal and urban geometries.	(*16*, *18*)
TUFLOW	Coupled 1D Pipe/ 2D Overland Hydrodynamic Model	The Two-dimensional Unsteady Flow (TUFLOW) modeling system is utilized as a coupled one-dimensional (1D) pipe and two-dimensional (2D) overland hydrodynamic model designed to simulate complex flow patterns and interactions in urbanized environments. TUFLOW solves the full 2D depth-average momentum and continuity equations for shallow water free surface flow while simultaneously incorporating the full functionality of the 1D hydrodynamic network to represent fluvial and pipe networks. The system dynamically links	(*18*)

		the 1D and 2D domains to simulate the exchange of water between the land surface and the underground drainage system.	
CityCAT 2D	2D Hydrodynamic Flood Model	The City Catchment Analysis Tool (CityCAT) is a 2D hydrodynamic flood model that operates based on the evaluation of infiltration of pervious areas. To generate the underlying topography, the model utilizes Digital Terrain Models (DTMs) as inputs, while also employing topographic spatial data to describe the features of the built environment. By including parameters such as buildings and soil porosity, CityCAT allows for a more realistic simulation of flow paths in urban areas. Recent applications of the model have leveraged Cloud computing to simulate a large number of ensembles, allowing researchers to assess the uncertainty and variability of extreme rain events under both present and future conditions. The final output of these simulations consists of spatial footprints of the flood event, producing a series of hazard maps that show water depths.	(*80*)
FVCOM	Hydrodynamic Model	Finite Volume Community Ocean Model (FVCOM) is an unstructured-grid, finite-volume ocean model used for simulating circulation and transport processes in coastal and estuarine environments. It is a core component of operational flood forecasting systems like the Northeast Coastal Ocean Forecast System (NECOFS), which provides real-time predictions of water levels that are critical for emergency management and protecting transportation infrastructure during storm events.	(*16*)

Hydrodynamic models are indispensable tools for assessing coastal flood impacts on transportation infrastructure. Their primary strength lies in simulating the complex physical processes of compound events, such as the interaction between upstream-propagating storm surge and downstream river discharge, which is a principal threat to low-lying transportation corridors (*16*, *34*, *37*, *73*). The outputs of these models—detailed maps of flood extent, depth, and duration—provide the foundational data for comprehensive vulnerability assessments (*52*, *53*, *57*, *110*). This hazard information is used not only to identify physically exposed infrastructure but also, when integrated with travel demand models, to quantify system-wide cascading consequences such as traffic delays and economic disruption (*25*, *52*, *53*, *82*). Therefore, the application of these advanced models is a fundamental first step in developing robust, evidence-based adaptation strategies to enhance the resilience of critical transportation infrastructure (*1*, *18*).

Network Analysis and Accessibility-Based Models

While hydrodynamic models are crucial for identifying which transportation links are physically exposed to flooding, they do not inherently quantify the resulting degradation of the entire network's performance (*50*, *53*, *80*, *97*). To bridge this gap, researchers represent the transportation system as a network graph of nodes and links, often using GIS to overlay flood maps and analyze network topology (*8*, *52*, *80*). Common methods include Accessibility Analysis, which measures the ease of reaching destinations, and Network Centrality Assessment (NCA), which uses graph theory metrics like Betweenness and Closeness Centrality to identify the most critical road segments (*8*, *12*, *50*). By simulating the removal of flooded links, analysts can quantify consequences like increased travel time and lost trips, as demonstrated in a Boston case study where such impacts doubled under projected climate conditions (*53*).

To achieve a more nuanced impact assessment than a simple binary (flooded/not-flooded) approach, some studies develop depth-disruption functions that establish an empirical relationship between floodwater depth and the corresponding reduction in vehicle speed (*30*, *88*). A study in Bangalore, for instance, applied this concept to show that a 1-in-100-year event would render over 40% of the road network immovable (*88*). Ultimately, this integrated analysis framework is crucial for identifying critical infrastructure where failure would cause the most severe cascading effects, thereby informing the prioritization of climate change adaptation interventions and investments to enhance overall network resilience (*9*, *28*, *29*, *52*, *57*).

Statistical and Machine Learning Approaches

The increasing frequency of coastal flooding poses a significant threat to urban resilience and critical transportation infrastructure (*7*, *16*, *32*, *48–50*, *52*, *53*, *57*, *80*, *81*, *97*). Accurate risk assessment requires sophisticated statistical and machine learning (ML) methodologies to quantify flood hazards and their interaction with the built environment (*7*, *25*, *71*, *111*).

Statistical and Probabilistic Modeling

Statistical methods are vital for developing a probabilistic understanding of flood risk and managing uncertainty in projections (*21*, *36*, *48*, *63*). Techniques such as Monte Carlo analysis inform infrastructure defense spending (*21*, *36*, *48*, *53*, *73*, *91*), while Extreme Value Analysis (EVA) is standard for estimating rare event magnitudes to set design standards (*13*, *16*, *23*, *34*, *48*, *54*, *91*, *105*). Because coastal flooding often results from the interaction of multiple drivers (e.g., rainfall and storm surge), assuming these factors are independent can significantly underestimate risk (*7*, *15*, *16*, *19*, *21*, *32*, *48*, *51*, *52*, *54*, *56*, *71*, *83*, *97*). Consequently, copula functions are widely

used to model the joint probability of these drivers, enabling a more accurate calculation of compound flood return periods, especially as climate phenomena like ENSO can alter these dependencies (*15*, *16*, *21*, *21*, *54*, *73*, *77*, *97*).

Machine Learning Approaches

Machine learning offers powerful data-driven tools for flood assessment, starting with susceptibility mapping, which is especially valuable in data-scarce regions (*6*, *7*, *32*, *63*, *71*, *97*). Supervised algorithms like AdaBoost, Logistic Regression, and Maximum Entropy use conditioning factors (e.g., elevation, land use) to generate high-accuracy hazard maps, which can then be integrated with MCDA methods like EWM-TOPSIS to produce comprehensive risk maps incorporating vulnerability indicators such as road density (*7*, *32*, *63*, *71*, *97*, *105*). For more immediate applications, ensemble methods like Bayesian Model Combination (BMC) integrate learners such as Backpropagation Neural Networks (BPNN) and Random Forests (RF) with IoT sensor data for near-real-time forecasting (*6*, *97*). Complementing this, Artificial Neural Networks (ANNs) serve as computationally efficient surrogate models to provide rapid inundation forecasts based on complex hydrodynamic model outputs (*16*, *97*, *112*).

By integrating these statistical and ML methodologies, researchers can develop a holistic understanding of compound flood risk, providing a quantitative foundation for designing resilient transportation infrastructure (*49*, *52*, *80*, *97*).

Integrated Frameworks

Assessing coastal flood impacts on transportation infrastructure requires integrated frameworks that move beyond simple inundation mapping to analyze functional performance and cascading societal effects (*9*, *16*, *33*, *53*)s. These state-of-the-art methodologies synthesize diverse models and data sources to create a holistic understanding of risk. The most sophisticated approaches can be categorized by their method of integration, including the coupling of hydrodynamic and network models, the integration of probabilistic and physical models, and the combination of data-driven machine learning with knowledge-based multicriteria analysis.

Coupling Hydrodynamic Models with Network Analysis

A primary integrated approach links hydrodynamic flood models with network analysis platforms to simulate how the physical failure of an asset propagates through an interdependent system, causing cascading failures (*9*, *25*). For instance, one such framework couples a 2D hydrodynamic model (LISFLOOD-FP) with the RiskLogik platform to model how an initial flood event can disrupt interconnected transportation, energy, and communication sectors (*9*, *56*, *110*). This coupling can also quantify the functional performance of the transportation system itself (*18*, *33*, *53*, *80*). By integrating flood maps with an activity-based travel demand model (e.g., MATSim), researchers can evaluate how roadway damage affects individual travel behavior and quantifies system-wide consequences, such as the total increase in travel time during peak hours (*33*, *53*, *80*). Other studies have used coupled models to measure flood-induced accessibility reductions across roadway networks, providing a direct metric of transportation service disruption (*18*, *27*, *28*, *33*, *53*).

Integrating Probabilistic Methods and Physical Models

To manage uncertainty in future climate projections, advanced frameworks integrate probabilistic methods with physical models, using statistical outputs to define hazard scenarios

(*21*, *63*, *91*, *97*). At a continental scale, this involves using an ensemble of Global Circulation Models (GCMs) and large catalogs of synthetically generated hurricanes to drive a high-resolution hydrodynamic model that simulates fluvial, pluvial, and coastal hazards (*19*). At regional and local scales, EVA of historical data is used to define low-frequency events (e.g., 100-year return period) that serve as boundary conditions for coupled 1D-2D hydrodynamic models (*3*, *34*, *37*, *52*, *56*, *91*, *105*, *110*). Similarly, probability density functions (PDFs) can be fitted to historical data to estimate the 100-year wave setup and storm surge, which are then combined with probabilistic SLR projections to calculate an extreme total water level for hazard mapping (*3*, *37*).

Combining GIS, MCDA, and ML

Another powerful integration combines data-driven ML with the knowledge-based frameworks of GIS and MCDA (*32*, *97*). In this approach, ML algorithms, such as an AdaBoost-Logistic model, are first trained on historical flood data and conditioning factors (e.g., elevation, land use) to produce a highly accurate flood hazard map (*32*, *63*, *97*). This hazard layer is then integrated within a GIS environment with various socioeconomic vulnerability layers, such as population and poverty density (*32*, *52*, *55*, *81*). The relative importance of these vulnerability factors is determined by using an MCDA technique like the AHP, which assigns weights to each layer (*32*, *97*). The combination of the hazard and weighted vulnerability maps produces a final, holistic flood risk assessment that is a practical tool for planners to identify and prioritize at-risk regions (*32*, *52*, *63*, *105*, *110*).

These integrated frameworks represent the cutting edge of the field, enabling a more robust and nuanced understanding of flood risk that accounts for physical processes, systemic interdependencies, and the deep uncertainties inherent in projecting future hazards (*97*).

RESEARCH GAPS AND OPPORTUNITIES

While research into the impacts of coastal flooding on transportation systems has grown significantly, a comprehensive review of existing literature reveals several critical gaps and corresponding opportunities for future inquiry. To establish a roadmap for subsequent studies, this review maps identified research gaps directly to the conceptual framework established in Section 2. A quantitative analysis of the 83 reviewed studies reveals distinct disparities in how drivers are characterized, how physical damage is modeled, and which transportation modes are prioritized.

Gaps in Driver Characterization: The Subsidence Deficit

While literatures are increasingly recognizing the complexity of flood hazards, a significant portion of research remains fragmented. Among the studies that specified a flood driver, 40 papers focused on single flood drivers in isolation—primarily storm surge (17), sea level rise (13), or rainfall (10)—ignoring the nonlinear amplification that occurs during interactions. Although 37 studies addressed compound events, they did not consider all the possible drivers of compound flooding. Additionally, there is a critical gap regarding Geophysical Preconditions. Specifically, VLM and subsidence were explicitly modeled as primary drivers in only 4 papers. This omission is particularly alarming for deltaic regions like the U.S. Gulf Coast, where the neglect of subsidence rates leads to a systematic underestimation of future relative sea-level rise.

Gaps in Impact Analysis: Methodological Fidelity

A major methodological disconnect exists between flood exposure modeling and structural engineering requirements. Of the studies that detailed their modeling methodology, 34 utilized

hydrodynamic modeling capable of simulating physics-based flow. However, 24 studies still relied on simplified, static approaches, including (13), bathtub models (7), and map-based assessments (4). Notably, methodology information was unspecified or not applicable in 25 of the total reviewed papers, often due to a focus on qualitative policy frameworks rather than quantitative engineering analysis. This reliance on static inundation depth in over a third of the quantitative literature is problematic because it neglects the hydrodynamic forces (velocity, drag, lift) that are often the primary cause of structural failure.

System-Level Gaps: The Modal Bias

The most pronounced gap identified in this review is the overwhelming modal bias toward roadway networks. Among the 44 studies that focused on a specific transportation mode, 31 concentrated exclusively on roads or road networks. In contrast, critical logistical nodes are severely underrepresented, with rail systems (3), ports (1), and airports (1) receiving minimal attention. This scarcity extends to the fundamental mechanics of vulnerability; unlike roadway networks, where depth-disruption functions are relatively well-established, there is a critical absence of validated damage curves for non-road modes. Consequently, specific failure thresholds—such as the hydrodynamic load required to trigger railway ballast washout or the inundation depth that necessitates port terminal shutdowns—remain largely undefined. Furthermore, nearly half of the total reviewed papers did not specify a single mode, often analyzing critical infrastructure as a generic aggregate. This lack of specificity and the low adoption of Multimodal approaches obscure the systemic risks inherent in supply chains, where a failure in a seaport can cascade to affect connecting rail and trucking networks.

Methodological Gaps: Addressing Deep Uncertainty

Finally, current planning methodologies remain predominantly deterministic. While machine learning approaches are emerging, the vast majority of studies rely on fixed climate scenarios that may not capture the full range of deep uncertainty associated with future climate projections. There is a clear need to progress toward dynamic, probabilistic frameworks—such as Dynamic Adaptive Policy Pathways (DAPP)—that allow for flexible decision-making. Additionally, the integration of social equity metrics into technical risk assessments remains underdeveloped, with few studies quantifying how transport disruptions disproportionately affect disadvantaged populations during evacuation and recovery (*55*, *63*).

In summary, future research must progress beyond static, deterministic, and isolated analyses. The most significant opportunities lie in advancing a more integrated and holistic paradigm. This involves developing coupled, probabilistic models that capture dynamic system responses; adopting a network-based perspective to analyze multimodal systems and their interdependencies; and creating sophisticated, multi-dimensional vulnerability metrics that account for functional, social, and economic factors.

CONCLUSION

This systematic review synthesizes state-of-the-art research regarding the impacts of coastal flooding on transportation infrastructure through an assessment of 83 peer-reviewed studies. The analysis demonstrates that the field is undergoing a critical paradigm shift, evolving from static, single-hazard assessments toward a more integrated characterization of compound climate events. A principal conclusion is that traditional risk assessments frequently underestimate vulnerability by analyzing flood drivers in isolation. By aligning the literature with the IPCC AR6

typology, this review elucidates that the most severe impacts arise not from passive inundation, but from multivariate compound and preconditioned events, where non-climatic factors—specifically VLM and subsidence—significantly amplify relative sea-level rise. The frequent omission of subsidence data, particularly within deltaic regions such as the U.S. Gulf Coast, constitutes a critical deficiency in current predictive modeling.

Furthermore, this review identifies a substantial disconnect between hydraulic modeling and structural performance analysis. While static inundation models remain prevalent, empirical evidence from extreme events, such as Hurricane Katrina, demonstrates that physical failure is frequently governed by hydrodynamic forces—specifically drag, lift, and wave slam—rather than submergence depth alone. Consequently, methodologies that fail to couple hydraulic velocity with structural physics prove inadequate for accurately predicting asset degradation.

From a systemic perspective, the review reveals a pronounced modal bias in existing literature, with 70% of research concentrating exclusively on roadway networks. This road-centric focus marginalizes critical logistical nodes—such as seaports, airports, and rail systems—thereby obscuring the cascading failures that propagate through multi-modal supply chains. Although this review was limited by the disciplinary fragmentation of the literature and a scope restricted to English-language publications (2000–2025), the rigorous methodology employed ensures a robust assessment of current research capabilities.

These findings delineate a clear trajectory for future studies: research must progress beyond deterministic, single-mode analyses. The field necessitates the development of dynamic, probabilistic modeling frameworks that capture the nonlinear interactions of compound drivers, integrate real-time subsidence data, and simulate the functional interdependencies of the entire transportation network. By identifying these specific mechanistic and systemic gaps, this review serves as a foundational resource to direct future inquiry and support evidence-based strategies for the development of resilient coastal infrastructure.

FUNDING SOURCE

This study is funded by the U.S. Department of Transportation through the Coastal REsearch And Transportation Education - CREATE University Transportation Center under Grant Award Number 69A3552348330.

AUTHOR CONTRIBUTIONS

The authors confirm contribution to the paper as follows: Md Shah Mominul Islam Momin: conceptualization, methodology, draft and final manuscript preparation, and result visualization; Eunsang Cho: project supervision, conceptualization, and manuscript review/editing; Eunsaem Cho: methodology and manuscript review/editing; Subasish Das: conceptualization, data guidance, and manuscript review/editing. All authors reviewed the results and approved the final version of the manuscript.

DECLARATION OF CONFLICTING INTERESTS

Subasish Das is a member of Transportation Research Record's Editorial Board. All other authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

ACKNOWLEDGEMENTS

Portions of this manuscript were edited for language clarity, grammar, and stylistic consistency using a large language model (Gemini). The tool was used solely to improve readability and organization of text drafted by the authors and was not used for data collection, data analysis, model development, interpretation of results, or generation of figures or tables. In addition, the Zotero reference management software was used to organize references, manage citations, and ensure consistent formatting of in-text citations and the reference list.

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